Problem 4: Drone Rain
7 Points

Problem ID: `rain`
Rank: `2`

Introduction

California is in a drought again! With water bills rising and the rainy season approaching, Alex Sharc and his environmental activism startup Plant-ir have decided to unleash a fleet of drones designed to collect ~~surveillance data~~ rainwater to reduce water bills for California residents.

Problem Statement

Plant-ir sets up a list $\mathbf{L}$ of $\mathbf{N}$ drones in three-dimensional space numbered from `1` to $\mathbf{N}$. Each drone $\mathbf{L_i}$ starts at initial coordinates $(\mathbf{X}_i, \mathbf{Y}_i, \mathbf{Z}_i)$ and each day moves one unit in a fixed direction $\mathbf{D_i}$, either `E` (east) or `N` (north). The $\mathbf{X}_i$ and $\mathbf{Y}_i$ coordinates represent the drone's position on the horizontal plane, while the $\mathbf{Z}_i$ coordinate represents its height above the plane. If a drone moves `E` (east) it moves **right** along the x-axis. If a drone moves `N` (north) it moves **up** along the y-axis.

The drones operate over $\mathbf{K}$ consecutive nights. On each night, the following happens:

- **Rainfall:** each drone $\mathbf{L_i}$ collects **1 unit of rainwater** as long as there is no drone directly above it (i.e., there is no drone $\mathbf{L_j}$ such that $\mathbf{X}_i = \mathbf{X}_j$, $\mathbf{Y}_i = \mathbf{Y}_j$, and $\mathbf{Z}_i < \mathbf{Z}_j$)
- **Movement:** each drone moves one unit in its assigned direction $\mathbf{D_i}$ while maintaining a constant height $\mathbf{Z}_i$

It is guaranteed that no two drones initially share both $\mathbf{X}$ and $\mathbf{Y}$ coordinates and that no two drones will ever collide in three-dimensional space (same $\mathbf{X}$, $\mathbf{Y}$, and $\mathbf{Z}$). Determine the total amount of rainwater collected by all drones over all $\mathbf{K}$ nights.

Note: Rain falls immediately on the first night, before any drones begin moving.

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Input Format

The first line of input contains a single integer $\mathbf{T}$, denoting the number of test cases. For each test case:

- The first line contains two space-separated integers $\mathbf{N}$ and $\mathbf{K}$, where:
  - $\mathbf{N}$ denotes the number of drones.
  - $\mathbf{K}$ denotes the number of nights.
- The following $\mathbf{N}$ lines four space-separated integers $\mathbf{X}_i$ $\mathbf{Y}_i$ $\mathbf{Z}_i$ $\mathbf{D}_i$, where:
  - $\mathbf{X_i}$ is an integer denoting the starting x-coordinate of the drone $\mathbf{L_i}$.
  - $\mathbf{Y_i}$ is an integer denoting the starting y-coordinate of the drone $\mathbf{L_i}$.
  - $\mathbf{Z_i}$ is an integer denoting the height of the drone $\mathbf{L_i}$.
  - $\mathbf{D_i}$ is the character `N` or `E` denoting the direction of the drone $\mathbf{L_i}$.

Output Format

For each test case, output a single integer representing the total amount of rainwater collected by all the drones over all $\mathbf{K}$ nights.

Constraints

$1 \leq \mathbf{T} \leq 100$
$1 \leq \mathbf{N} \leq 10^5$
$1 \leq \mathbf{K} \leq 10^9$
$0 \leq \mathbf{X}_i, \mathbf{Y}_i, \mathbf{Z}_i \leq 10^5$

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Sample Test Cases

Sample Input Download

```
3
3 3
0 3 0 E
2 1 1 N
8 8 2 E
4 2
0 5 0 E
5 0 1 N
1 3 2 E
3 1 3 N
5 5
0 6 0 E
2 4 1 E
4 2 2 N
5 1 3 N
9 9 4 N
```

Sample Output Download

```
8
8
22
```

Sample Explanation

For the first test case: on night 1, there are no drones on top of each other, so the drones collect 3 units of rainwater in total. On night 2, the drones are still on distinct coordinates, so they accumulate another 3 units of rainwater. On night 3, drone 1 and drone 2 both move to coordinate $(2, 3)$ on the x-y plane, causing drone 1 not to collect any water as drone 2 has a higher $\mathbf{Z}$ coordinate. Thus, the drones gain another 2 units of rainwater on night 3. In total, the drones collect $3 + 3 + 2 = 8$ units of rainwater.

For the second test case: the drones never meet at the same coordinates in the x-y plane. On each of the two nights, the drones collect 4 units of rainwater. In total, the drones collect $4 + 4 = 8$ units of rainwater.

For the third test case: on each of nights 1 and 2, the drones collect 5 units of rainwater. On each of nights 3, 4, and 5, the drones collect 4 units of rainwater. In total, the drones collect $5 + 5 + 4 + 4 + 4 = 22$ units of rainwater.

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